BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `c:\Users\sarah\OneDrive\Documents\GitHub\hw2-sarah-nathan`
   Installed Fontconfig_jll         v2.17.1+0
   Installed Libmount_jll          v2.41.1+0
   Installed JpegTurbo_jll          v3.1.3+0
   Installed InvertedIndices         v1.3.1
   Installed InlineStrings          v1.4.5
   Installed Accessors            v0.1.42
   Installed Roots              v2.2.10
   Installed DataFrames            v1.8.0
   Installed SentinelArrays          v1.4.8
   Installed WorkerUtilities         v1.6.1
   Installed WeakRefStrings          v1.4.2
   Installed StatsBase            v0.34.6
   Installed DataStructures          v0.19.1
   Installed TableTraits           v1.0.1
   Installed Expat_jll            v2.7.1+0
   Installed PooledArrays           v1.4.3
   Installed DataValueInterfaces       v1.0.0
   Installed Plots              v1.40.19
   Installed Ghostscript_jll         v9.55.1+0
   Installed CompositionsBase         v0.1.2
   Installed PrettyTables           v3.0.8
   Installed CommonSolve           v0.2.4
   Installed InverseFunctions         v0.1.17
   Installed Libuuid_jll           v2.41.1+0
   Installed Crayons             v4.1.1
   Installed OpenSSL_jll           v3.5.2+0
   Installed Tables              v1.12.1
   Installed CSV               v0.10.15
   Installed Latexify             v0.16.10
   Installed Glib_jll             v2.86.0+0
   Installed IteratorInterfaceExtensions   v1.0.0
   Installed FilePathsBase          v0.9.24
   Installed StringManipulation       v0.4.1
   Installed SortingAlgorithms        v1.2.2
Precompiling project...
   3770.4 ms   IteratorInterfaceExtensions
   2767.0 ms   DataValueInterfaces
   2858.0 ms   WorkerUtilities
   2806.4 ms   CommonSolve
   2842.7 ms   InverseFunctions
   2721.0 ms   CompositionsBase
   2732.6 ms   InvertedIndices
```

```
 2818.2 ms    InlineStrings
 3473.9 ms    Crayons
 1715.6 ms    PooledArrays
 1907.0 ms    JLLWrappers
 5767.9 ms    SentinelArrays
 5994.4 ms    DataStructures
 4242.7 ms    FilePathsBase
 5496.4 ms    RecipesBase
 5452.6 ms    StringManipulation
 5920.8 ms    Unitful → ConstructionBaseUnitfulExt
 3954.9 ms    TableTraits
 1251.8 ms    CompositionsBase → CompositionsBaseInverseFunctionsExt
 1720.4 ms    Graphite2_jll
 1902.2 ms    Unitful → InverseFunctionsUnitfulExt
 2534.3 ms    InverseFunctions → InverseFunctionsDatesExt
 2952.1 ms    LogExpFunctions → LogExpFunctionsInverseFunctionsExt
 1641.0 ms    Libmount_jll
 2835.5 ms    OpenSSL_jll
 3534.6 ms    InverseFunctions → InverseFunctionsTestExt
 1757.9 ms    EpollShim_jll
 1806.4 ms    LLVMOpenMP_jll
 1796.5 ms    Bzip2_jll
 1738.6 ms    Xorg_libXau_jll
 1782.6 ms    Xorg_libICE_jll
 1771.1 ms    libpng_jll
 1811.6 ms    libfdk_aac_jll
 1842.8 ms    LAME_jll
 1815.2 ms    LERC_jll
 1543.9 ms    fzf_jll
 1532.8 ms    Ogg_jll
 1695.2 ms    JpegTurbo_jll
 1864.4 ms    XZ_jll
 1534.6 ms    mtdev_jll
 1437.2 ms    Xorg_libXdmcp_jll
 1751.5 ms    x265_jll
 1615.7 ms    x264_jll
13178.3 ms    ColorSchemes
 1654.3 ms    libaom_jll
 1673.7 ms    Zstd_jll
 1698.0 ms    Expat_jll
 1357.7 ms    libevdev_jll
 1586.3 ms    LZO_jll
 1574.8 ms    Opus_jll
```

```
 1396.8 ms    Xorg_xtrans_jll
 1430.4 ms    Libiconv_jll
 1332.1 ms    eudev_jll
 1502.5 ms    Libffi_jll
 1494.7 ms    Libuuid_jll
 1464.7 ms    SortingAlgorithms
 2016.0 ms    FriBidi_jll
 2498.8 ms    FilePathsBase → FilePathsBaseMmapExt
 2885.1 ms    Tables
 3473.7 ms    FilePathsBase → FilePathsBaseTestExt
18675.2 ms    Parsers
 1692.4 ms    Xorg_libSM_jll
 2809.3 ms    FreeType2_jll
 2701.6 ms    libvorbis_jll
 1944.1 ms    Xorg_libxcb_jll
 3226.2 ms    Pixman_jll
 3368.6 ms    JLFzf
 3960.8 ms    Ghostscript_jll
 5438.0 ms    OpenSSL
 3674.7 ms    Dbus_jll
 3582.7 ms    libinput_jll
 7916.6 ms    Accessors
 3589.6 ms    Wayland_jll
 3724.1 ms    GettextRuntime_jll
 4693.4 ms    Libtiff_jll
 2233.2 ms    Xorg_xcb_util_jll
 2146.1 ms    Xorg_libX11_jll
 3978.7 ms    Accessors → LinearAlgebraExt
 6989.4 ms    JSON
 7117.9 ms    InlineStrings → ParsersExt
 8035.4 ms    Fontconfig_jll
 2688.4 ms    Accessors → UnitfulExt
 2895.2 ms    Accessors → TestExt
 2263.6 ms    Xorg_xcb_util_image_jll
 1923.8 ms    Xorg_xcb_util_renderutil_jll
 2169.9 ms    Xorg_xcb_util_keysyms_jll
 4927.5 ms    Glib_jll
 1978.9 ms    Xorg_xcb_util_wm_jll
12154.6 ms    Latexify
16836.8 ms    StatsBase
 1830.3 ms    Xorg_libXrender_jll
 1822.5 ms    Xorg_libXext_jll
 1769.8 ms    Xorg_libXfixes_jll
```

```
 2143.2 ms  Xorg_libxkbfile_jll
 3352.2 ms  Xorg_xcb_util_cursor_jll
 6945.2 ms  Xorg_libXinerama_jll
 7767.3 ms  WeakRefStrings
 9182.2 ms  UnitfulLatexify
11422.6 ms  Roots
 9155.6 ms  Latexify → SparseArraysExt
29790.9 ms  PlotUtils
 5500.2 ms  Xorg_libXrandr_jll
 3028.8 ms  Libglvnd_jll
 1968.3 ms  Xorg_libXcursor_jll
 2045.3 ms  Xorg_libXi_jll
 2024.9 ms  Xorg_xkbcomp_jll
 3688.5 ms  Roots → RootsUnitfulExt
 1689.8 ms  Xorg_xkeyboard_config_jll
 6632.7 ms  Cairo_jll
 1603.0 ms  xkbcommon_jll
 1960.5 ms  Vulkan_Loader_jll
 5545.6 ms  HarfBuzz_jll
11915.8 ms  PlotThemes
 6032.2 ms  Qt6Base_jll
 6179.1 ms  Pango_jll
 6871.9 ms  libass_jll
16734.5 ms  RecipesPipeline
 1905.0 ms  libdecor_jll
 2017.7 ms  GLFW_jll
45080.1 ms  HTTP
17035.7 ms  Qt6ShaderTools_jll
16212.0 ms  FFMPEG_jll
64733.0 ms  PrettyTables
 7904.9 ms  Qt6Declarative_jll
 4933.3 ms  FFMPEG
49383.9 ms  CSV
10423.6 ms  Qt6Wayland_jll
15021.0 ms  GR_jll
 6353.5 ms  GR
67414.5 ms  DataFrames
 7427.1 ms  Latexify → DataFramesExt
95265.6 ms  Plots
 6714.3 ms  Plots → UnitfulExt
133 dependencies successfully precompiled in 227 seconds. 86 already precompiled.
```

```julia
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

> **Tip**
>
> Think about Henry's law for CO2.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

> **Tip**
>
> Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system in compliance with a regulatory limit of 2.3 kg/(1000m^3). You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized "Snowball Earth". In this problem, we'll introduce a simple model of the Earth's energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} \underbrace{+ a \ln\left(\frac{[CO_2]}{[CO_2]_{PI}}\right)}_{\text{greenhouse effect}}, \tag{1}$$

where:

- T is the Earth's global mean temperature (in $^\circ$C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m^2/$^\circ$C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m^2/$^\circ$C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m^2 (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

Problem 3.2

Rather than assuming a constant value for the albedo α, we will represent the ice-albedo feedback by letting α depend on T:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T+10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoprotereozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

References

List any external references consulted, including classmates.